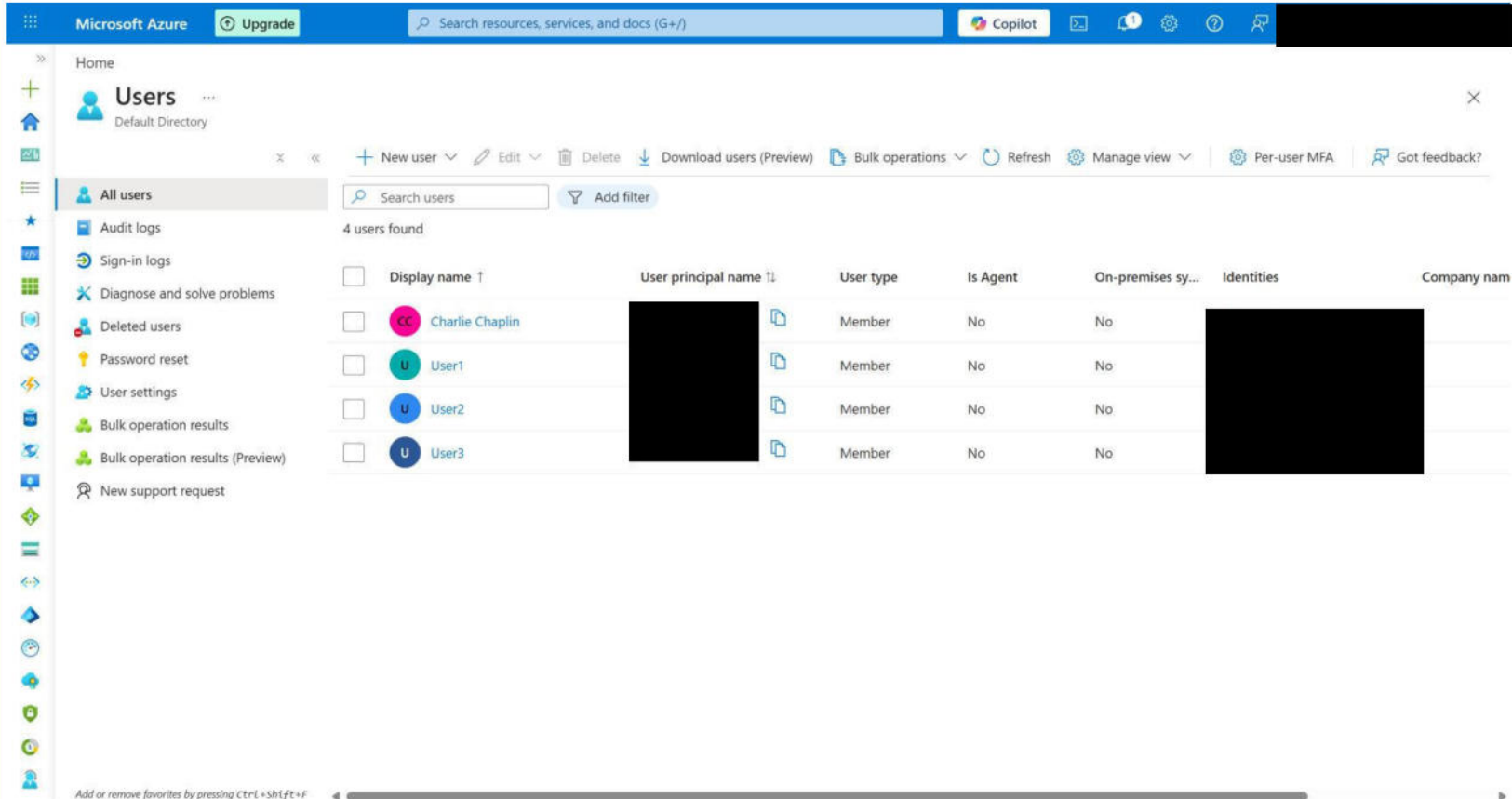


Azure Security Capstone Project Solution

Instructions: Document the screenshots required as indicated below.

Task 1: Create a group with three users, assign the Virtual Machine Administrator Login role, and implement multifactor authentication for the users.

Activity	Key evidence																																			
Activity 1: Create three users in Azure Active Directory (Azure AD).	Screenshot that shows the three users on the Users page  <table><tr><th>Display name ↑</th><th>User principal name ↑</th><th>User type</th><th>Is Agent</th><th>On-premises sy...</th><th>Identities</th><th>Company nam</th></tr><tr><td>☐ Charlie Chaplin</td><td>[Redacted]</td><td>Member</td><td>No</td><td>No</td><td>[Redacted]</td><td></td></tr><tr><td>☐ User1</td><td>[Redacted]</td><td>Member</td><td>No</td><td>No</td><td>[Redacted]</td><td></td></tr><tr><td>☐ User2</td><td>[Redacted]</td><td>Member</td><td>No</td><td>No</td><td>[Redacted]</td><td></td></tr><tr><td>☐ User3</td><td>[Redacted]</td><td>Member</td><td>No</td><td>No</td><td>[Redacted]</td><td></td></tr></table>	Display name ↑	User principal name ↑	User type	Is Agent	On-premises sy...	Identities	Company nam	☐ Charlie Chaplin	[Redacted]	Member	No	No	[Redacted]		☐ User1	[Redacted]	Member	No	No	[Redacted]		☐ User2	[Redacted]	Member	No	No	[Redacted]		☐ User3	[Redacted]	Member	No	No	[Redacted]	
Display name ↑	User principal name ↑	User type	Is Agent	On-premises sy...	Identities	Company nam																														
☐ Charlie Chaplin	[Redacted]	Member	No	No	[Redacted]																															
☐ User1	[Redacted]	Member	No	No	[Redacted]																															
☐ User2	[Redacted]	Member	No	No	[Redacted]																															
☐ User3	[Redacted]	Member	No	No	[Redacted]																															

Activity 2: Create a group for the three users and assign the Virtual Machine Administrator Login role to this group.

1) Screenshot that shows the three users on the **All groups** page

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

Home > Groups | All groups > JewelSales

JewelSales | Members

Group

+ Add members

Bulk operations

Refresh

Manage view

Remove

Got feedback?

Overview

Diagnose and solve problems

Manage

Properties

Members

Owners

Roles and administrators

Administrative units

Group memberships

Applications

Licenses

Azure role assignments

Activity

Troubleshooting + Support

Direct members

All members

Search

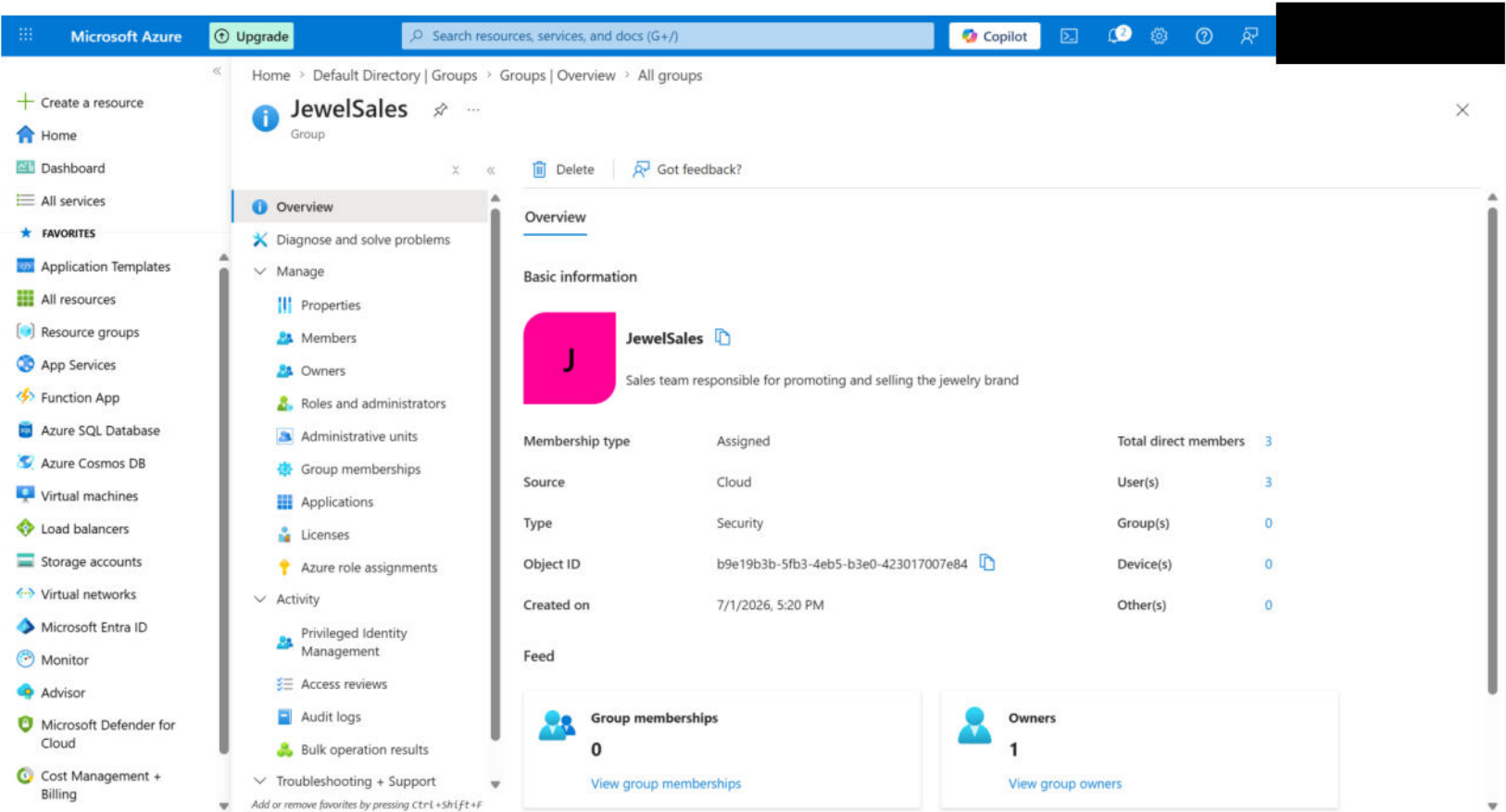
Add filter

3 group members found

	Name ↑	Type	Email	User type	Object ID
☐	 User1	User		Member	cc6b112
☐	 User2	User		Member	6cb0f63
☐	 User3	User		Member	1095246

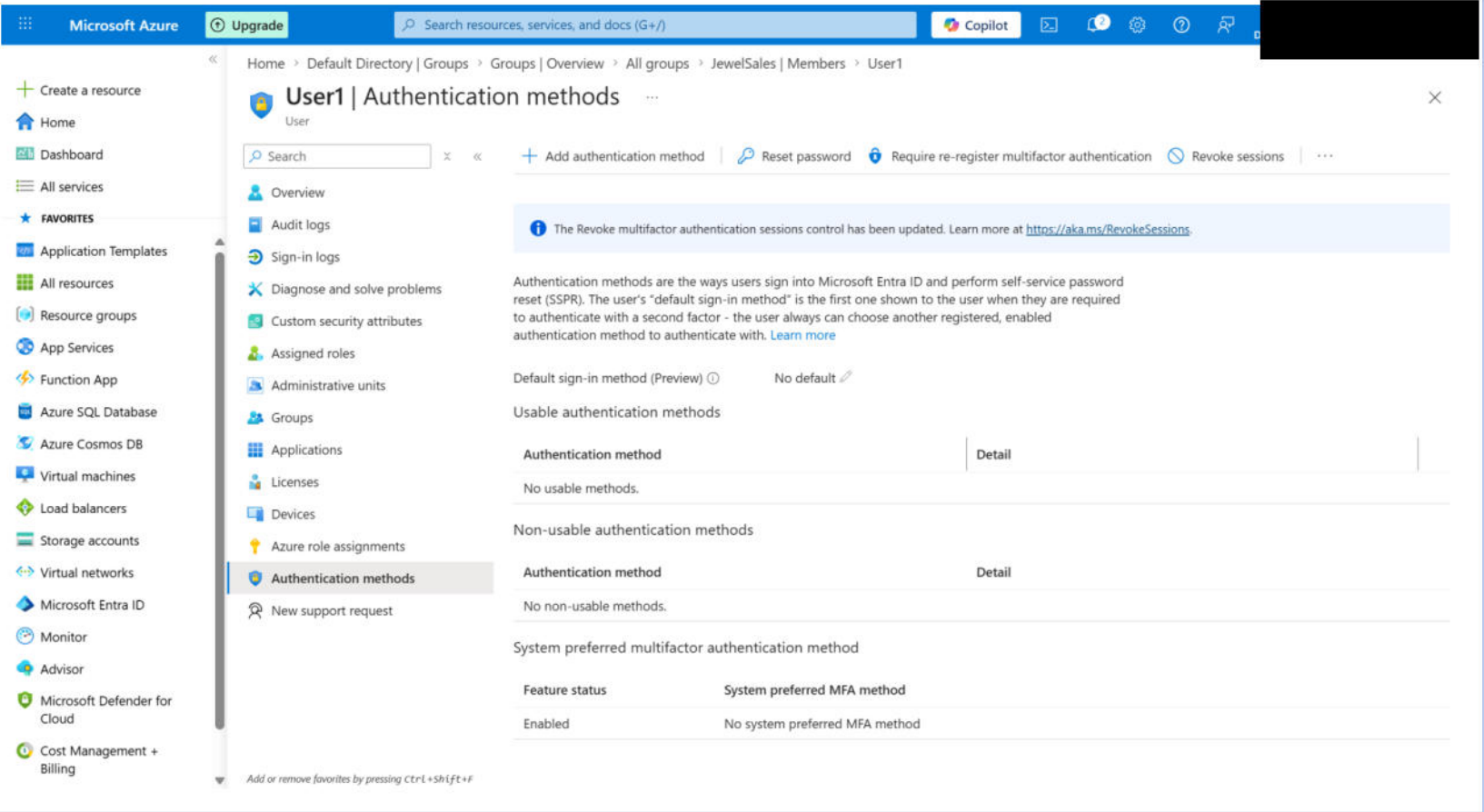
Add or remove favorites by pressing Ctrl+Shift+F

2) Screenshot that shows the **Added Role assignment** notification indicating that the group was assigned the Azure Virtual Machine Administrator Login role (cannot be done as I don't have Entra P1/P2 license)

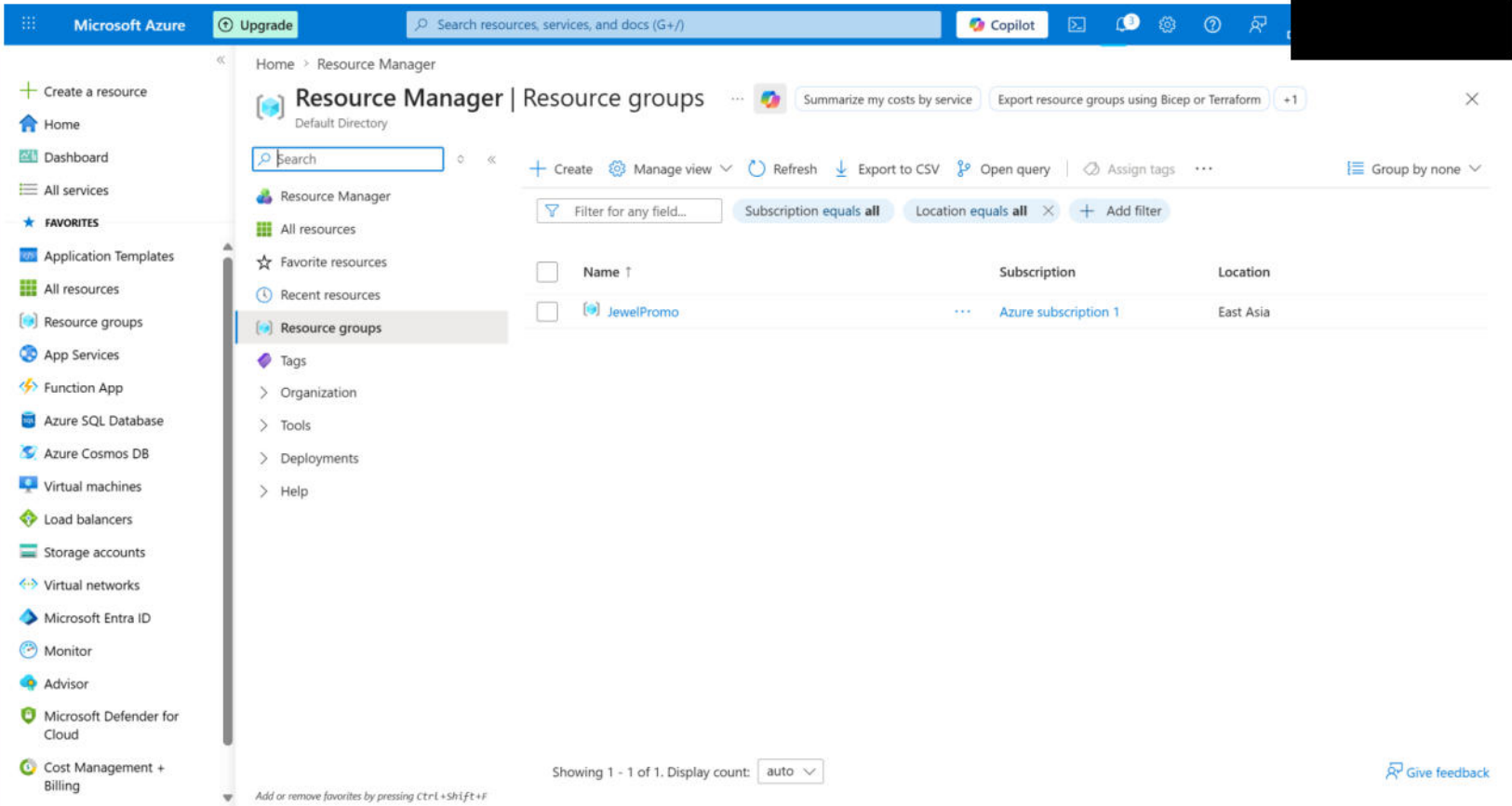


Activity 3: Implement mandatory multifactor authentication for all three users.

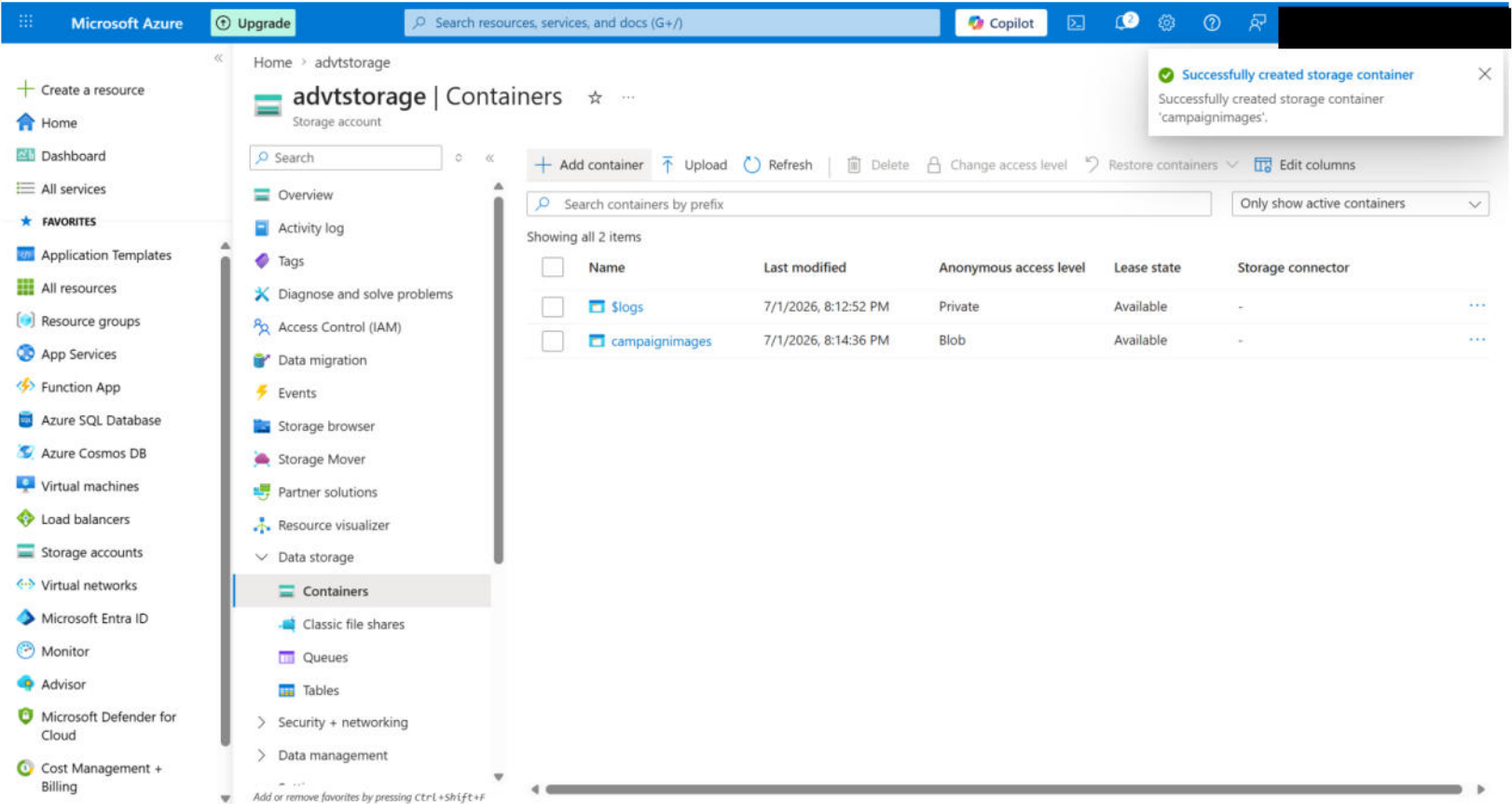
Screenshot that shows the **MULTI-FACTOR AUTH STATUS** enabled for all three users. MFA is enabled by default for all users.



Task 2: Upload an image to a Storage account container, create a Blob shared access signature (SAS) URL for the image, and assign the Storage Blob Data Contributor role to the group members.

Activity	Key evidence
Activity 1: Create a resource group, a Storage account, and a container.	1) Screenshot that shows the new resource group listed on the Resource groups page 

2) Screenshot that shows the Containers page of the new resource group with the new container listed



Activity 2: Upload an image to the container in the Storage account and create a Blob SAS URL for the image.

1) Screenshot that shows the uploaded image on the Storage account's **Containers** page

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > advtstorage | Containers

campaignimages

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Successfully uploaded blob(s)

Successfully uploaded 6 blob(s).

+ Add Directory

Upload

Change access level

Refresh

Delete

Copy

Paste

Rename

Acquire lease

campaignimages

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

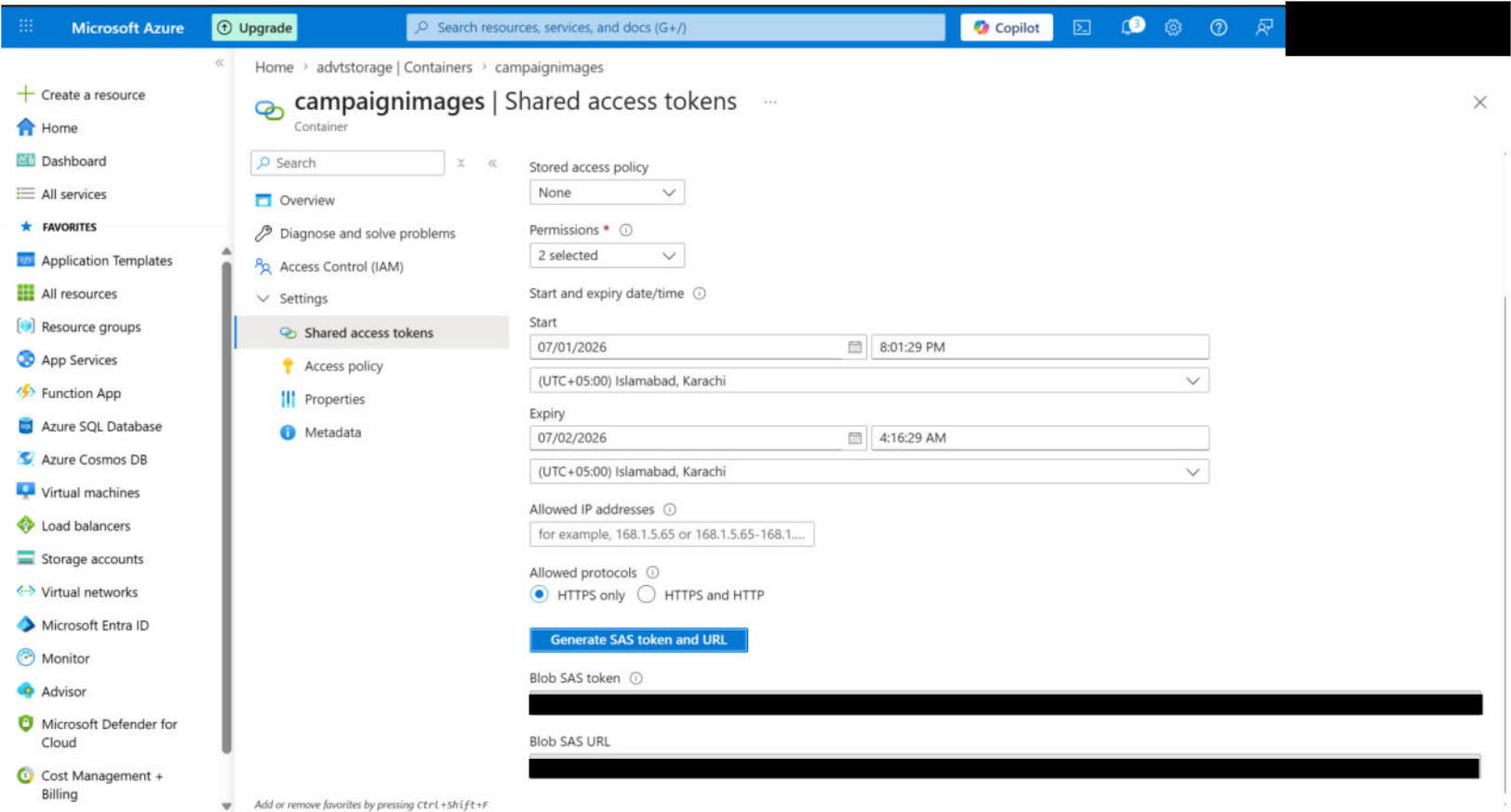
Only show active blobs

Showing all 6 items

	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	10.webp	7/1/2026, 8:15:47 PM	Hot (Inferred)	Block blob	23.33 KiB	Available	...
☐	5.webp	7/1/2026, 8:15:47 PM	Hot (Inferred)	Block blob	175.03 KiB	Available	...
☐	6.webp	7/1/2026, 8:15:47 PM	Hot (Inferred)	Block blob	27.8 KiB	Available	...
☐	7.webp	7/1/2026, 8:15:47 PM	Hot (Inferred)	Block blob	17.81 KiB	Available	...
☐	8.webp	7/1/2026, 8:15:47 PM	Hot (Inferred)	Block blob	19.15 KiB	Available	...
☐	9.webp	7/1/2026, 8:15:47 PM	Hot (Inferred)	Block blob	23.6 KiB	Available	...

Add or remove favorites by pressing Ctrl+Shift+F

2) Screenshot that shows the Blob SAS URL for the image



Activity 3: Assign the Storage Blob Data Contributor role to the group members.

Screenshot that shows the **Storage Blob Data Contributor** role assigned to the three users on the **Access control (IAM)** page of **Resource groups**

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > advtstorage | Containers > campaignimages

campaignimages

Container

Search

Access Control (IAM)

Overview

Diagnose and solve problems

Settings

Shared access tokens

Access policy

Properties

Metadata

Access Control (IAM)

Owner (2)

Charlie Chaplin 2f043ba1-7fc9-4aa... User Owner Subscription (Inherite... None

Charlie Chaplin 2f043ba1-7fc9-4aa... User Owner Subscription (Inherite... None

Microsoft Sentinel Contributor (2)

Microsoft Threat ... 8ee8fdad-f234-42... Service principal Microsoft Sentinel Contri... Subscription (Inherite... None

WindowsDefende... fc780465-2017-4... Service principal Microsoft Sentinel Contri... Subscription (Inherite... None

Storage Blob Data Contributor (3)

User1 cc6b112e-bcda-40... User Storage Blob Data Contri... This resource Add

User2 6cb0f633-b546-40... User Storage Blob Data Contri... This resource Add

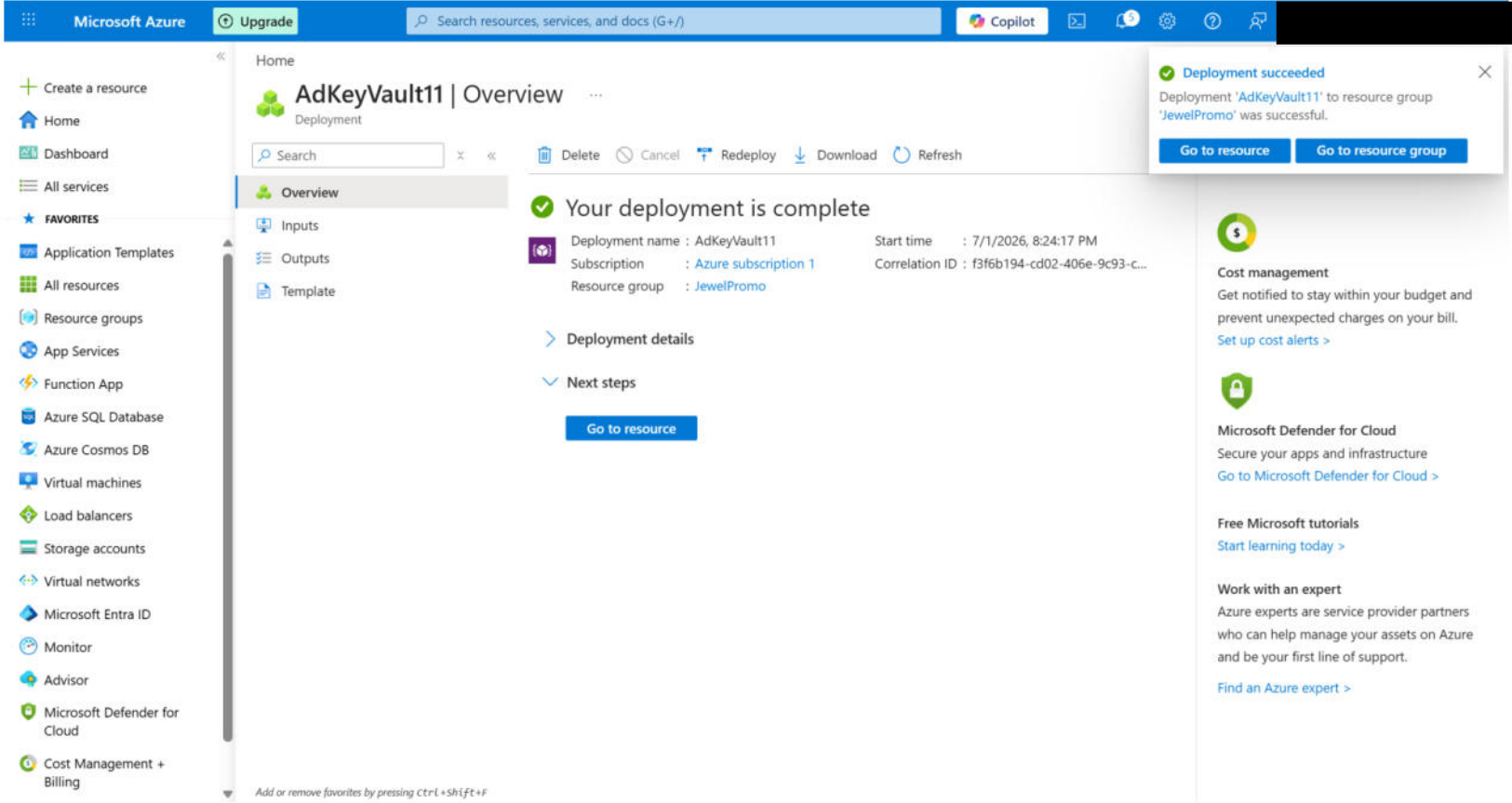
User3 10952400-ffa5-475... User Storage Blob Data Contri... This resource Add

User Access Administrator (1)

Charlie Chaplin 2f043ba1-7fc9-4aa... User User Access Administrator Root (Inherited) None

Showing 1 - 8 of 8 results.

Task 3: Create a Key Vault and a managed identity and encrypt the Storage account using customer-managed keys.

Activity	Key evidence
Activity 1: Create a Key Vault and a managed identity and assign key permissions to the managed identity.	1) Screenshot that shows a notification that the Key Vault is successfully deployed  The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and a 'Copilot' button. The left sidebar lists various services, with 'Key Vault' highlighted under 'All services'. The main content area shows the 'Overview' page for 'AdKeyVault11', indicating that the deployment is complete. A notification banner at the top right states 'Deployment succeeded' for the deployment 'AdKeyVault11' to resource group 'JewelPromo'. The deployment details show the name 'AdKeyVault11', subscription 'Azure subscription 1', and resource group 'JewelPromo'. The start time is 7/1/2026, 8:24:17 PM, and the correlation ID is f3f6b194-cd02-406e-9c93-c... The 'Next steps' section includes a 'Go to resource' button. The right sidebar contains links for 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'.

2) Screenshot that shows a notification that the managed identity is successfully deployed

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and icons for Copilot, help, and settings. The left sidebar lists various Azure services and favorites. The main content area shows the 'Overview' page for a deployment named 'CreateForm-20260701202510'. A green checkmark icon indicates that the deployment is complete. The deployment details show the name, subscription, resource group, start time, and correlation ID. A 'Go to resource' button is visible. A notification banner at the top right confirms the successful deployment. The right sidebar contains links to 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'.

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot

Home

CreateForm-20260701202510 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateForm-202607012025... Start time : 7/1/2026, 8:25:48 PM

Subscription : Azure subscription 1 Correlation ID : fdabaa59-d416-4ba3-80cd-...

Resource group : JewelPromo

> Deployment details

✓ Next steps

Go to resource

Deployment succeeded

Deployment 'CreateForm-20260701202510' to resource group 'JewelPromo' was successful.

Pin to dashbo... Go to resource gr...

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill. Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support. Find an Azure expert >

Add or remove favorites by pressing Ctrl+Shift+F

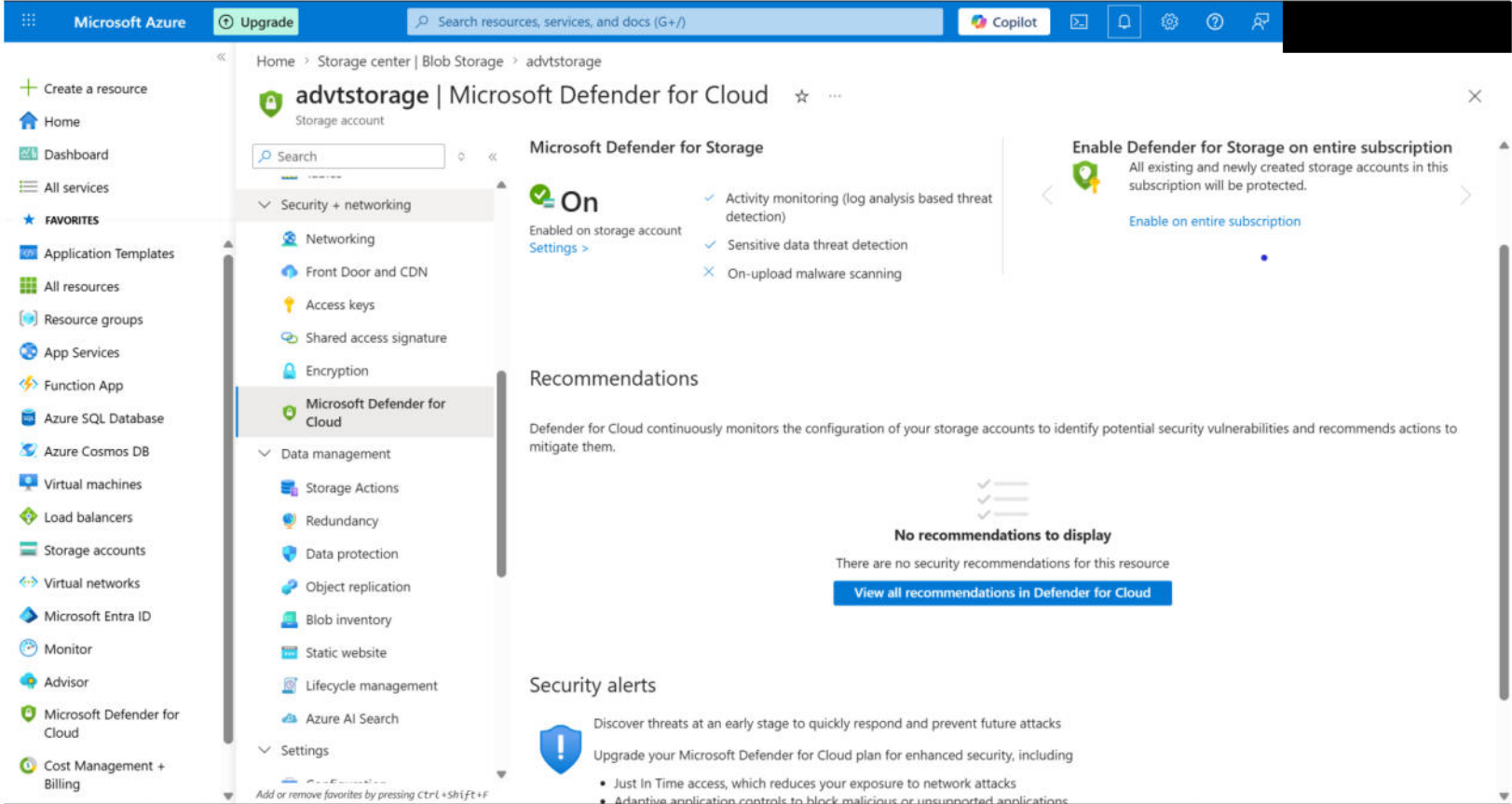
3) Screenshot that shows that key permissions are assigned to the managed identity

The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation options like 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area displays the 'Access control (IAM)' page for the Key Vault 'AdKeyVault11'. The page shows a list of role assignments with columns for Name, Type, Role, Scope, and Condition. The roles assigned include 'Owner' (2 users), 'Key Vault Administrator' (1 user), 'Key Vault Crypto User' (1 Managed identity), 'Microsoft Sentinel Contributor' (2 Service principals), and 'User Access Administrator' (1 user). The 'Key Vault Crypto User' role is assigned to the Managed identity 'CampaignIdentity'.

Name	Type	Role	Scope	Condition
Owner (2)				
Charlie Chaplin	User	Owner	Subscription (Inherited)	None
Charlie Chaplin	User	Owner	Subscription (Inherited)	None
Key Vault Administrator (1)				
Charlie Chaplin	User	Key Vault Administrator	This resource	None
Key Vault Crypto User (1)				
CampaignIdentity	Managed identity	Key Vault Crypto User	This resource	None
Microsoft Sentinel Contributor (2)				
Microsoft Threat ...	Service principal	Microsoft Sentinel Contri...	Subscription (Inherited)	None
WindowsDefende...	Service principal	Microsoft Sentinel Contri...	Subscription (Inherited)	None
User Access Administrator (1)				
Charlie Chaplin	User	User Access Administrator	Root (Inherited)	None

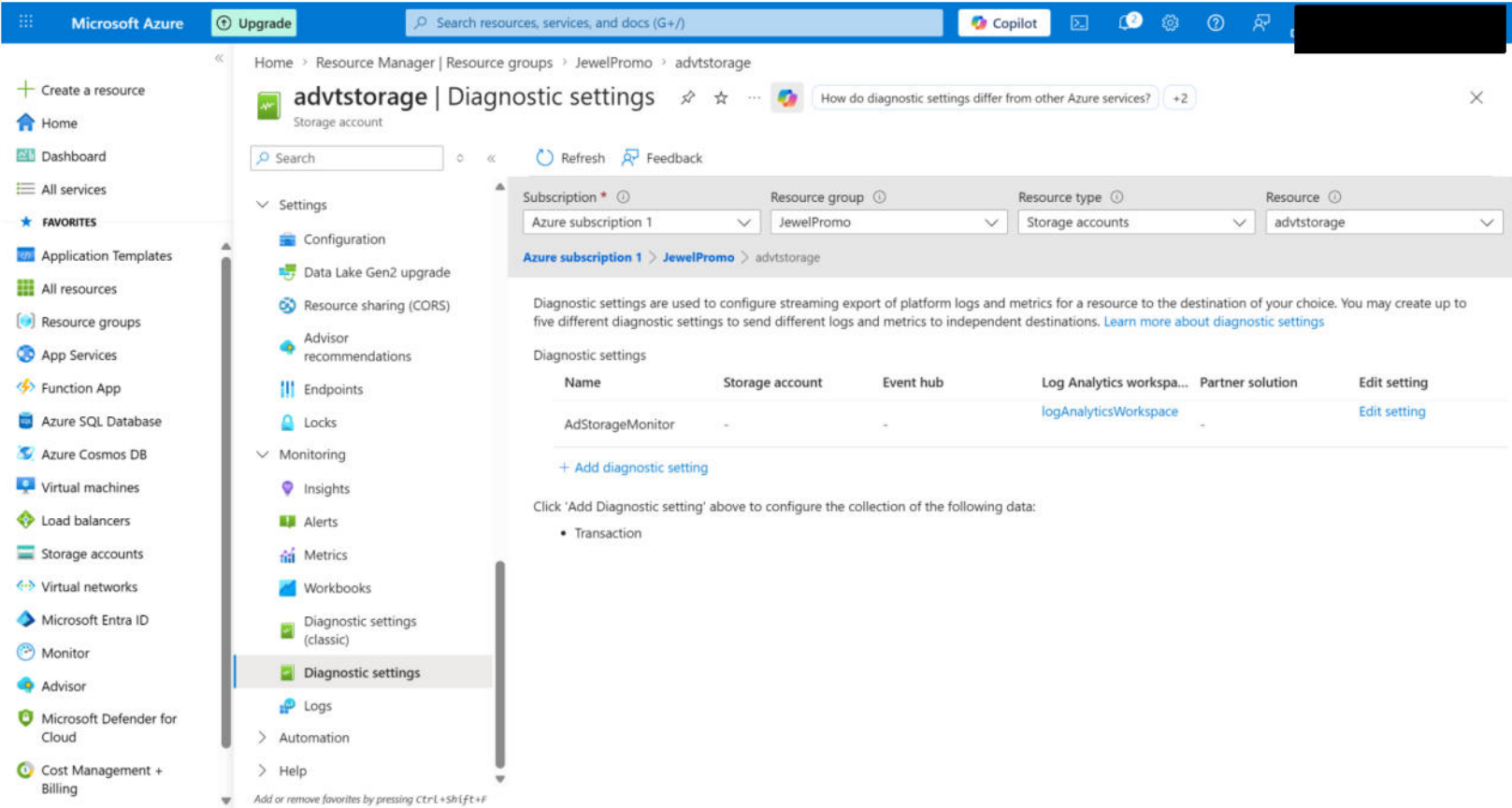
Showing 1 - 7 of 7 results.

Task 4: Secure your Storage account using Microsoft Defender and Log Analytics workspace.

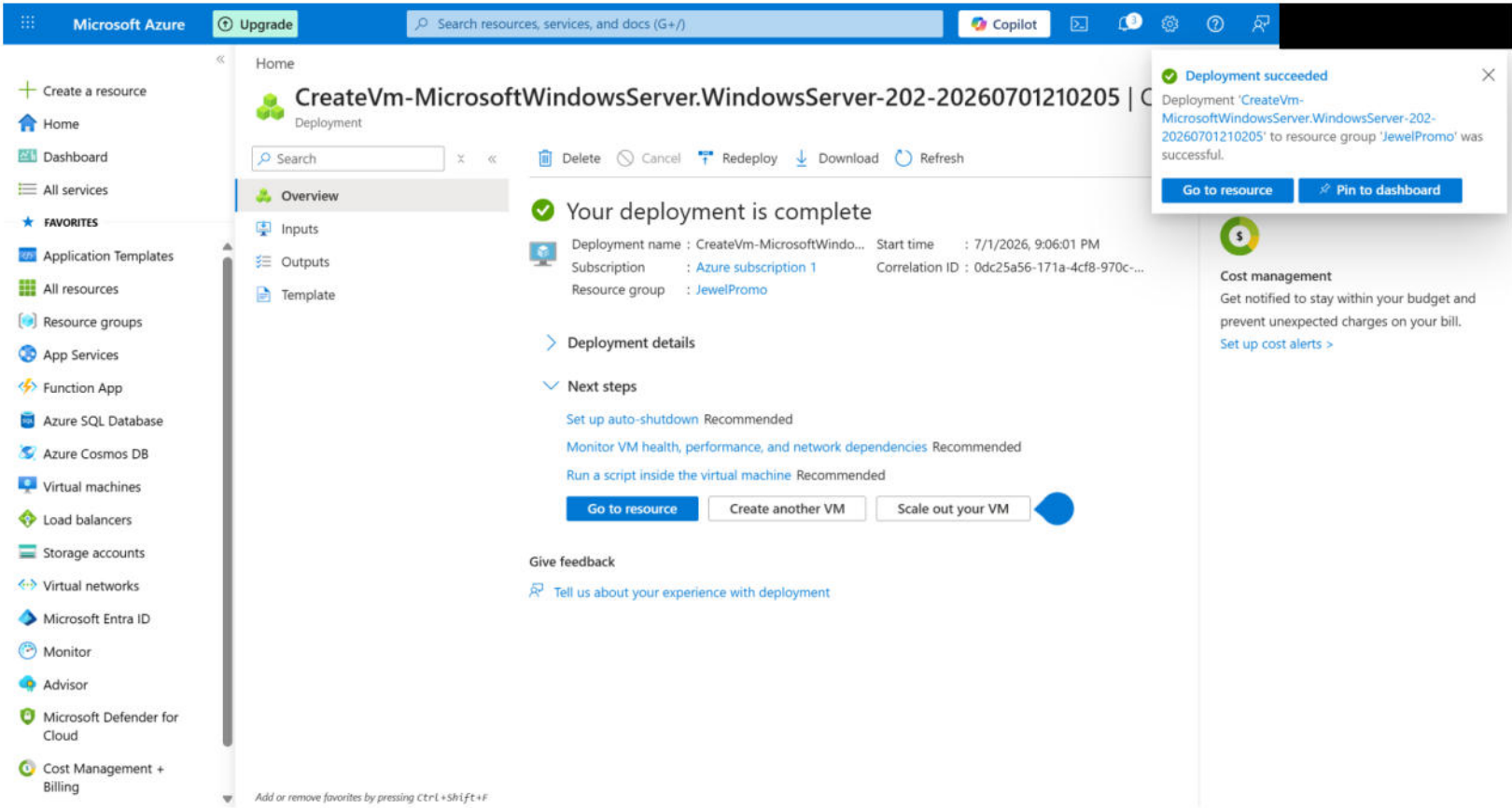
Activity	Key evidence
Activity 1: Enable Microsoft Defender for Storage for your Storage account.	Screenshot that shows the Microsoft Defender for Storage enablement status as On 

Activity 2: Configure diagnostic settings to send data to the Log Analytics workspace.

Screenshot that shows the new diagnostic setting on the **Diagnostic settings** page



Task 5: View the image using a Blob SAS URL on a virtual machine.

Activity	Key evidence
Activity 1: Create a Windows Server 2019 Datacenter virtual machine.	Screenshot that shows the Your deployment is complete message indicating your virtual machine has been created 

Activity 2: Add a network security group rule to allow inbound traffic on port 3389 for your virtual machine.

Screenshot that shows the new inbound port rule added for port 3389

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > Network foundation | Network security groups > CampaignVM-nsg

CampaignVM-nsg | Inbound security rules

Network security group

Search

+ Add

Hide default rules

Refresh

Delete

Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Inbound security rules

Outbound security rules

Network interfaces

Subnets

Properties

Locks

Monitoring

Automation

Help

Add or remove favorites by pressing Ctrl+Shift+F

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name

Port == all

Protocol == all

Source == all

Destination == all

Action == all

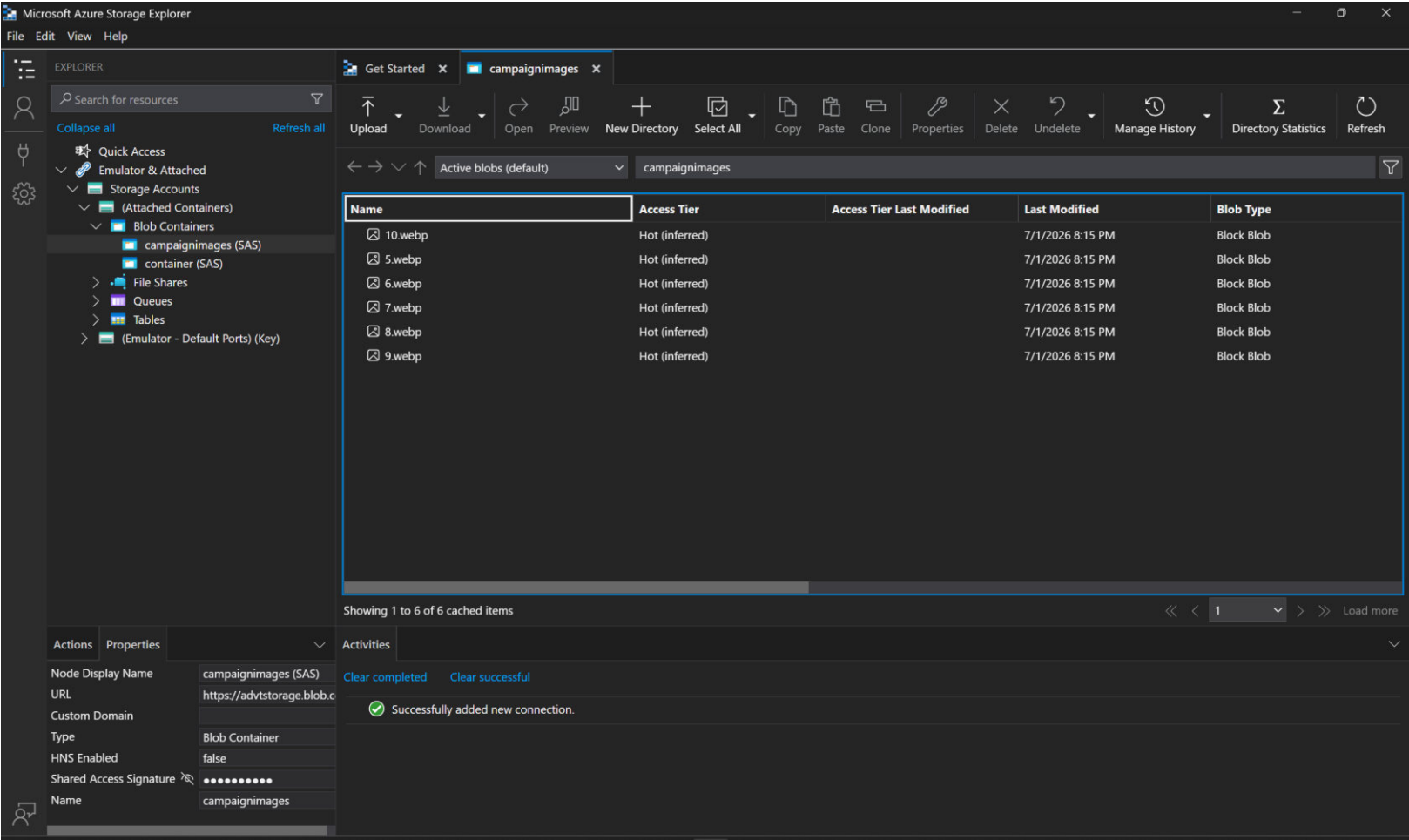
	Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓	Destination ↑↓	Ac
☐	310	AllowAnyCustom...	3389	Any	Any	Any	✓
☐	65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	✓
☐	65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any	✓
☐	65500	DenyAllInBound	Any	Any	Any	Any	✗

Created security rule

Successfully created security rule 'AllowAnyCustom3389Inbound'.

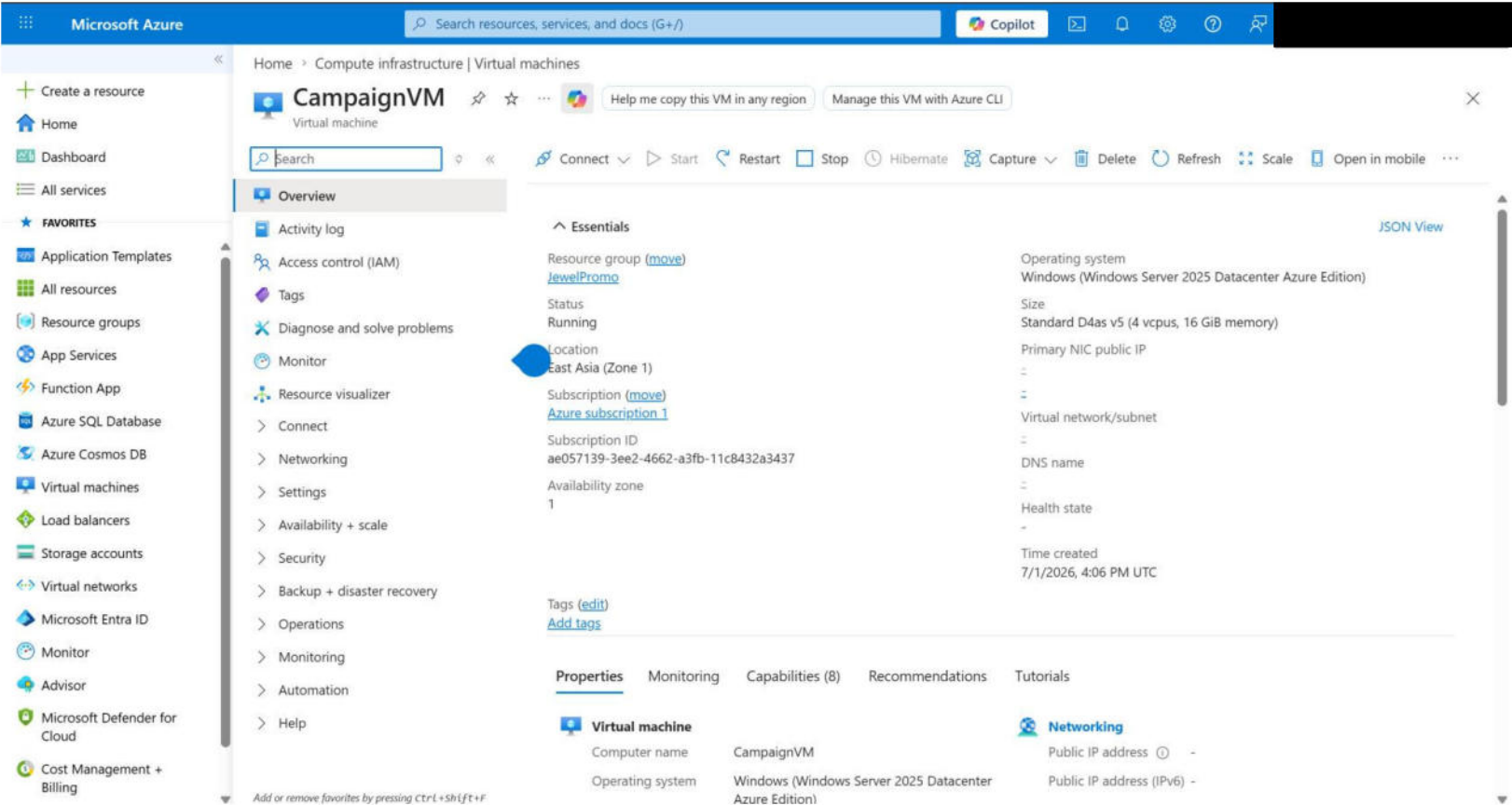
Activity 3: Verify access to the image as an administrator using the Blob SAS URL.

Screenshot that shows the images in MS Azure Storage Explorer with the Blob SAS URL

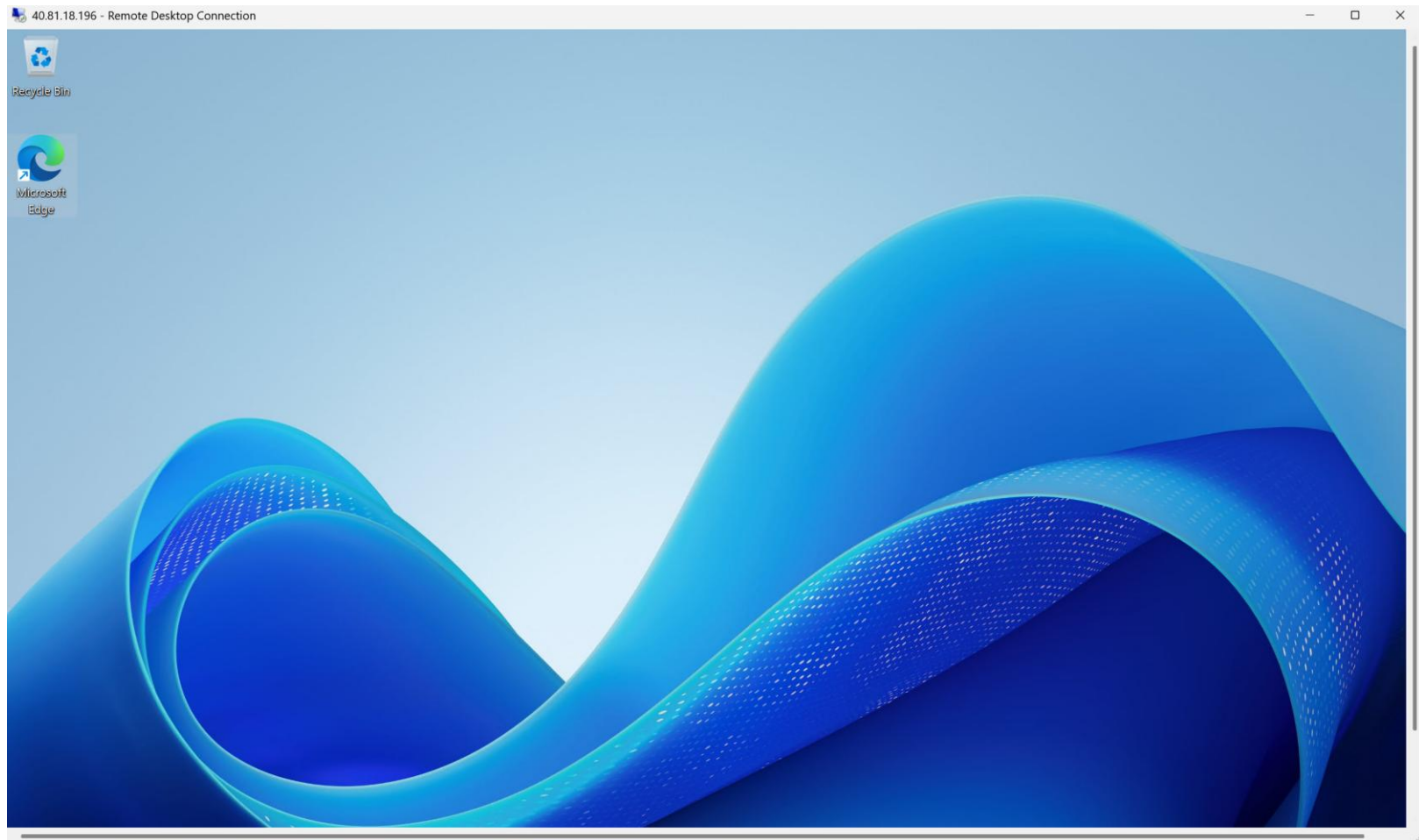


Activity 4: Access the image as a sales team member using the Blob SAS URL.

1) Screenshot that shows the Connect page of the virtual machine with one of the sales team members logged in



2) Screenshot that shows the image in a virtual machine browser with the Blob SAS URL



3) Final image of all the resources together in the same resource group.

The screenshot displays the Microsoft Azure portal interface for the 'JewelPromo' resource group. The left sidebar shows navigation options like 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area shows the 'Overview' tab for the resource group, listing various resources under the 'Essentials' section. The resources are displayed in a table with columns for Name, Type, and Location.

Name	Type	Location
AdKeyVault11	Key vault	East Asia
advstorage	Storage account	East Asia
CampaignIdentity	Managed Identity	East Asia
CampaignVM	Virtual machine	East Asia
CampaignVM-ip	Public IP address	East Asia
CampaignVM-nsg	Network security group	East Asia
CampaignVM-vnet	Virtual network	East Asia
campaignvm998_z1	Network Interface	East Asia
CampaignVM_OsDisk_1_a22a6ef75dee408f928f8aec437cb16e	Disk	East Asia
logAnalyticsWorkspace	Log Analytics workspace	East Asia

Showing 1 - 10 of 10. Display count: auto